By the end of the 1800s, physicists could see the outlines of a complete theory of the world coming into focus. Matter was made of atoms, which were made of smaller particles, interacting via various forces carried by fields, all operating under the umbrella of classical mechanics.

## What the World Is Made Of (Nineteenth-Century Edition)

- Particles (point-like, making up matter).
- Fields (pervading space, giving rise to forces).

New particles and forces would be discovered over the course of the twentieth century, but in the year 1899 it wouldn't have been crazy to think that the basic picture was under control. The quantum revolution lurked just around the corner, largely unsuspected.

If you've read anything about quantum mechanics before, you've probably heard the question "Is an electron a particle, or a wave?" The answer is: "It's a wave, but when we look at (that is, measure) that wave, it looks like a particle." That's the fundamental novelty of quantum mechanics. There is only one kind of thing, the quantum wave function, but when observed under the right circumstances it appears particle-like to us.

## What the World Is Made Of (Twentieth Century and Beyond)

- A quantum wave function.

It took a number of conceptual breakthroughs to go from the nineteenth-century picture of the world (classical particles and classical fields) to the twentieth-century synthesis (a single quantum wave function). The story of how particles and fields are different aspects of the same underlying thing is one of the underappreciated successes of the quest for unification in physics.

To get there, early twentieth-century physicists needed to appreciate two things: fields (like electromagnetism) can behave in particle-like ways, and particles (like electrons) can behave in wave-like ways.

The particle-like behavior of fields was appreciated first. Any particle with an electrical charge, such as an electron, creates an electric field